

Datum Fieldnotes: easing the data work burden on civic and non-profit organizations

William Eickman*
Georgia Institute of Technology
Atlanta, Georgia, USA
weickman3@gatech.edu

Mukhlisabonu K Nematova*
Wellesley College
Wellesley, Massachusetts, USA
mn109@wellesley.edu

Annabel Rothschild
Georgia Institute of Technology
Atlanta, Georgia, USA
arothschild@gatech.edu

Carl DiSalvo
Georgia Institute of Technology
Atlanta, Georgia, USA
cdisalvo@gatech.edu

Betsy DiSalvo
School of Interactive Computing
Georgia Institute of Technology
Atlanta, Georgia, USA
bdisalvo@cc.gatech.edu

Abstract

We present a Google Sheets add-on, Datum Fieldnotes, to enhance and formalize civic and non-profit (CNP) data workers' rich dataset annotation and contextualization practices. Unlike the massive, poorly-contextualized datasets typically used to train AI systems—which result in unexpected and harmful downstream system behavior—CNP datasets are usually crafted with an intimate understanding of the subject matter by the data workers creating and contributing to them, whether professionals, volunteers, or CNP clients. As part of a larger project, in this paper we report on the development of the Datum Fieldnotes tool and the results of user testing sessions with twelve CNP data workers. Our experience demonstrates how tools to facilitate data contextualization practices can be developed and supported. Following the user testing, we share a key revision to the tool: namely, the creation of a suite of training materials to not only ease tool onboarding for CNP data workers, but also to give them to the tools to interrogate and analyze the resultant paradata themselves, as a record of their own expertise and data labor contributed. This second contribution is critical to the development of CNP data work tools that are actually usable by practitioners in the long term.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**.

Keywords

Data annotation, data workers, AI ethics, civic and non-profit organizations

ACM Reference Format:

William Eickman, Mukhlisabonu K Nematova, Annabel Rothschild, Carl DiSalvo, and Betsy DiSalvo. 2025. Datum Fieldnotes: easing the data work burden on civic and non-profit organizations. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3706599.3719905>

1 Introduction

Civic and non-profit (CNP) organizations face a unique set of data work challenges. Their data work is mission critical, as it is needed both to track and demonstrate organizational efficiency, within the organization and to funders [21], yet it can take away time and resources from enacting the key mission of the organization: direct contact with clients [6]. Unlike data work that happens in for-profit settings, CNP data workplaces are often characterized by a lean budget and constant workforce fluctuations, as a result of volunteer labor and donation-dependent staffing capacity, leaving little time for training and on-the-job skill building [33]. At the same time, the data work that is done in CNP organizations is high-stakes; the ethical concerns related to doing related work properly (e.g., avoiding missing or mis-inputting data) originate in the organizational mission, which is directly concerned with improved the experience of daily life from clients from various populations in need, such as those experiencing homelessness [26]. At the same time, many commercial-grade data work platforms, such as CRMs [37], often require both expensive licensing fees and dedicated staff members to learn and maintain them [9, 15].

Traditional spreadsheet software has been identified in past literature as a more sustainable alternative data management software for CNPs, given the low barriers to entry and low (or no) cost license fees [29]. Spreadsheets, however, lack the advanced dataset documentation features needed to make actionable *paradata*, or process information about the development of a dataset that can be used for dataset contextualization [32]. Extensive prior work has shown that ethical design and analysis, as well as deployment of resultant systems, should be based on datasets that are well contextualized [13, 17, 19]. Paradata, as contextualizing material, is especially important for CNP data, which often focuses on improving human health and experience.

*Both authors contributed equally to this research.

Previous research in HCI has shown that CNP data workers already have robust dataset contextualization practices [6, 26, 28] that can result in workable paradata, *if* properly archived and shared. What is missing, however, are the tools to streamline and expedite this process, without adding to the heavy data work burden these data professionals already undertake. This is the purpose of Datum Fieldnotes, a Google Sheets add-on designed to help capture and formalize those dataset contextualization practices. Datum Fieldnotes¹ has two key functionalities. First, it automates micro-documentation of user changes per datum (or, cell-level) in the spreadsheet, which is missing from Google Sheets. Second, it provides an interface for users to create long-form notes, or records of questions, concerns, or observations about an individual datum value or datum that becomes embedded in the dataset itself. These functions are missing from Google Sheets' default environment; changes are queryable only by timestamp and comments do not reflect iterative timestamps or iterations. This results in workable paradata about a dataset; this record of genesis enables interrogation and contextualization of the dataset itself.

Datum Fieldnotes was developed as an artifact for a larger research-through-design (RtD) process, related to the development of tools, systems, and workflows for the development of better-contextualized data work. In this late breaking work, we focus on the design and testing of the tool itself, as a case study in the development of a data work tool specifically geared towards CNP workers, as an oft-neglected population of data workers. In particular, Datum Fieldnotes is designed to facilitate *data infrastructure literacy* for CNP workers and how design choices can support deeper dataset interrogation, bringing the background work of data work to the forefront [13].

In performing this work, we had two research questions. **RQ1**: can we offload some of the additional labor needed to document and archive the paradata that CNP data workers produce? And, subsequently, **RQ2**: how can we facilitate CNP data contextualization in action, to make sense of what other system features can help concretize this work? Our work addresses the urgent call to develop datasets that can build trust, reduce antagonism, help hold a mirror to organizational or governmental values, and to serve as a potential site of data contestation [19]. These practices are critical to developing responsible data-intensive systems, or ones produced off of datasets that are both well-understood and properly contextualized [24, 38]. Further, Datum Fieldnotes provides methods to centralize and record dataset authorship [12, 20], pointing to data workers as domain experts, who are best positioned to determine fair and appropriate use, in an era of rife data misuse [22].

2 Related Work

This work draws on two related areas: first, the data work burdens and challenges that CNPs undergo. Second, the utility of RtD as a method for studying data work in CNP organizations.

In referring to CNP data workers and data work, we combine two existing concepts. First, regarding civic data work/workers, we use Rothschild et al.'s [28] definition of civic data workers, which references those individuals who work on civic datasets, characterized by Sindera as data that reflects "people who live in the communities

and cities where the data was gathered" [30], while employed or volunteering for civic organizations. Similarly, non-profit data workers can be understood as those individuals who perform either compensated or volunteer data work while serving with a non-profit organization [34]. In both of these settings (civic and non-profit) the emphasis is on data work that directly supports or facilitates the mission of the organization, as opposed to a commercial product or service. In many cases, CNP data workers are domain experts in the organizational mission and secondarily perform data work such that it amplifies and extends that organizational mission.

The CNP Data Burden. CNPs rely on data work to demonstrate both organizational impact and potential to external funding bodies [34] but also to track progress towards internal goals (or benchmarks) [21] and identify relevant mission gaps or opportunities through open data [9]. However, there are several barriers for adoption of new data work tools or systems for CNPs [14]. There is high staff and volunteer turnover in many cases—due to grant-dependent positions and ad-hoc volunteering, respectively—which poses challenges for the maintenance of internal knowledge of data systems [15, 34]. When CNPs do have capacity for new data systems adoption, those systems are often developed by domain experts (in organizational mission) rather than data systems, resulting in what Volda et al. term *homebrew databases* [34]. Designated data manager staff members and advanced data work system and tool subscriptions count against the administrative budget or "overhead" on which CNPs are graded. Those grades are used by potential donors determine the organization's fiduciary prowess [25].

The emphasis on CNP's data can also re-direct attention from the organizational mission as a result of "data drift," where data findings push the mission in a new directions [3]. This creates an endless cycle of data work that indirectly supports the organization's mission (by helping accrue funding) [10], subsequently requiring more program evaluation data work [26]. Further, the burden of data work can also get in the way of the carrying out of the work directly related to the CNPs mission [6], for example, performing advocacy work or working directly with clients.

In short, improved systems must be flexible to the organizational mission, but require little customization, as the more nuanced or advanced a potential software solution is, the less likely it is to be usable for a CNP in the long term. This leaves CNPs in the lurch, resulting in data systems that are often rife with "fragmentation and duplication" which render them useless in the long-term, or even result in incorrect data analysis [2].

RtD for Studying CNP Data Work. Given the pressures on CNPs to produce data for funding bodies and for exchange with other related organizations, data work becomes a difficult task to tackle [36], in part, because little attention has been paid to CNP data workers, relative to their counterparts in commercial data science [28].

While these concerns have been documented by related work, there is a need to focus on applied methods for improving data practices in CNPs, both to support CNPs in getting the funding that they need [10] and to help them determine program efficacy and leverage existing open data resources [9]. We chose to employ the RtD method [31] in the larger project to which Datum Fieldnotes belongs because, to elicit the minutiae and nuance of CNP data work, we needed to see it happening in real time. Our goal is not

¹<https://dataworkforce.gatech.edu/datum-fieldnotes/>

only to understand CNP data work practices; our goal is to create something useful for our partner CNPs, which we elaborate on in the Discussion (5).

In particular, we wanted to know how we could support the data infrastructure literacies of our partner CNPs, including whether our partners needed support in understanding how a data system might align with the motivations for system design or dataset creation and use [21]. We employ a critical technical practice approach [1], aiming to assist CNPs in auditing their creation and use of their own datasets. Further, many CNP open datasets may be misused or misunderstood by downstream users—such as local governments trying to create new policy—and we want to help CNPs document and share out their data narrative or data stories, as they understand it [9]. Given that many CNP datasets end up as open datasets, these are increasingly subject to use by AI and ML system developers. Our goal is to enable CNPs to exert ownership over the data narratives that emerge and ensure that any predictive or generative system trained off these datasets matches the organizational mission of the CNPs that produced it.

3 Datum Fieldnotes Design

The first version of Datum Fieldnotes is based on the known needs of data workers in CNP organizations, based on our research team’s multi-year workplace ethnography with a data services provider, DataWorks, as well as known concerns raised by studies of CNP data work (see 2). DataWorks² provides contract labor for several CNP clients, focusing on early-stage data work. Drawing on this knowledge, we identified five critical, co-located implications for design resulting in the overall shape of the Datum Fieldnotes tool. The tool is 1) built on top of spreadsheet software, that is, 2) affordable, and, 3) easily accessible without vast technical background. It must allow for 4) dataset documentation to travel with the dataset itself and, 5) follow industry-grade data documentation practices.

Datum Fieldnotes: Spreadsheet Add-On for Google Sheets. CNP workers need data work tools that are accessible, adaptable, and replicable [29]. While Excel is commonly used for many businesses and organizations, we know that many CNP groups utilize Google Sheets, in part due to the collaboration-first nature of the service, and the lack of license fees (and time requirement for applying for license-fee waivers) [35]. Further, Google Sheets closely aligns with Shaprio et al.’s design tenets [29]. As Sheets is a cloud-first spreadsheet service, integrated with the existing Google Workplace Suite—already used by many CNPs—sharing access and security is handled by Google, rather than individual organizations [15]. Sheets already has a plethora of online resources and support unlike custom software suites. For those CNP data workers unfamiliar with Google Sheets, these existing learning materials minimize training overhead [18, 26].

Sidebar: Approachable functionality for data workers’ observations, reflections, and questions. In the design of Datum Fieldnotes, we prioritize user comfort with documentation input and attempt to automate the process of micro-documentation (rather than asking CNP workers to double their task load). The key input feature is the Notes functionality, housed on the Sidebar, while the Log (an additional tab added to the Sheets project on launching

Datum Fieldnotes) displays both those manual annotations, as well as a record of all individual data transactions. The Sidebar opens when a user opens the tool on a new sheet and is populated with up to five changes from that session (or since the user last opened the Sheets project). The recent displayed correspond to the cell that the user is currently interacting with, as the user performs a data transformation on any cell in the dataset. The Sidebar is a graphical user interface that sends information to the Log, enabling users to see and interact with information about recent spreadsheet changes. Notes allow users to create textual reflections on a particular cell or individual data transaction (cell change) to explain or question a given value, or flag an issue. Once a note is created, the cell to which the note is attached stays highlighted and is recorded to the Log.

Log: Automating micro-documentation in the background.

Upon first opening the tool, a separate tab in the sheet (workbook) is created labeled as the “Log”. For every edit made to a cell in the sheet while Datum Fieldnotes is active (including adding a Note), the Log will have an entry added with information about the change, including where the change was made, what the change was, when it was made, and who made it [10, 27]. This information is stored chronologically in the Log, which can be easily queried for a particular change, author, or timestamp, automating the collection of interrogatable *paradata* [32]. This change-by-change version history adds more specificity and detail than the standard version history that Google sheets keeps a record of automatically. As with any Sheets project, when the tab on which the tool is active is shared, the Log sheet is shared along with it. This means the dataset *paradata* can travel natively with the dataset itself, unlike a separate README document or datasheet. The Log is also a fully exportable as a CSV, enabling cross-platform dataset sharing and utilization [7]. The ability to share annotations recorded by the system better supports data sharing, a key requirement for CNP organizations [28]. For example, a housing non-profit data worker can now share their dataset (either as a Google Sheet or CSV) with city officials, contractors, or other actors in the same field *without* having to worry about compatibility issues.

Datasheets for Datasets integration: Cross-industry and downstream-aware documentation. There is an increasing movement to describe a dataset’s origin and intended use in the form of a datasheet, or documents shipped with hardware components to explain their capacities and capabilities [11]. Datasheets for datasets (D4D) have been shown to support data-intensive system designers in noticing and engaging with ethical issues related to dataset use [4]. Thus we incorporated several questions from D4D focusing on the development and intended use of the dataset in a new tab, which, like the Log, is added to the project when a user first launches Datum Fieldnotes. Including these D4D questions promotes further contextualization and reflection by data workers—who are experts in a dataset’s contents and formation—to serve as guides for downstream users or collaborators.

4 User Testing

To evaluate Datum Fieldnotes in context, we conducted 12 IRB-approved usability tests with participants who work with CNP data.

²For more information about DataWorks see <https://dataworkforce.gatech.edu/>

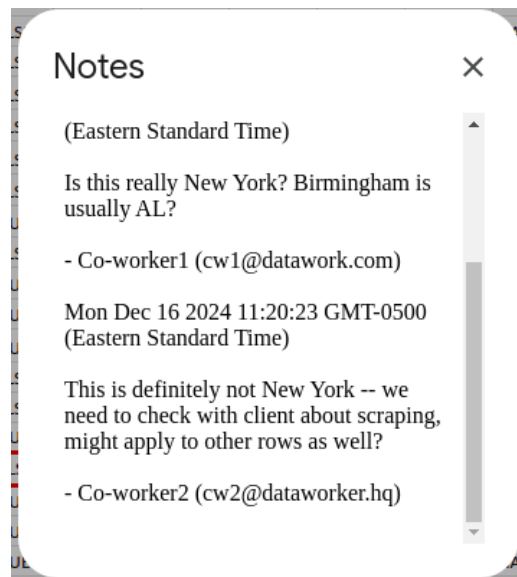


Figure 1: Notes screen showing iterative comments left on a single cell over time.

Methods. We recruited interviewees from both Tech Jobs and our own network of CNP community partners. When participants signed up to participate, we asked them to give a description of the kind of spreadsheet datasets they usually work with. We then used ChatGPT to create a synthetic dataset that emulated the genre of data (e.g., ecological measurements collected from sensors), so that participants were not required to share their own potentially sensitive organizational data. ChatGPT is certainly not without its own privacy and ecological concerns, however we chose to use it for two reasons. First, we wanted to avoid indirectly inflicting the research team’s prior knowledge of particulars of participants’ datasets, which we sometimes knew from prior collaboration and conversations about their work. Second, we hoped that ChatGPT’s propensity towards *AI mirages* [23], or faulty representations, would help create surprising or unusual data points that would pique participants’ curiosity. The research team then annotated the synthetic dataset as if we were participants’ colleagues, adding questions, comments, and changes on which we based a contextual inquiry activity [16]. In this activity participants traced these data points and their transformations, to make sense of their imaginary colleague’s concerns, reflections, or motivations in the data work they had conducted. Before this, participants were asked about their background with spreadsheet-based data work, in semi-structured interview format [8].

All participants described working with spreadsheets as a core competency of their work, from tracking numerical distribution of material resources for clients to thematic analysis of transcribed oral histories from community residents. The first seven participants were from DataWorks (referred to with the prefix ‘DW #’). These participants were not directly compensated; the study was voluntary and participants were indirectly compensated with their standard hourly wage (research engagement is a compensated part

of their roles). The other five participants are CNP practitioners (referenced as ‘P#’), working in fields from support for veterans with PTSD to eradicating hunger through food distribution. These participants were compensated with \$25 Amazon gift cards.

Testing took place via both in-person sessions and virtually via a recorded Zoom call. In all cases but one (resulting from machine incompatibility), the researcher’s desktop was used, to avoid unintentional exposure of participants’ data. The first three authors then conducted thematic analysis [5] of recorded sessions via their transcripts, notes taken during sessions, and post-session reflections. The resulting codebook was then applied to all corpus materials.

Participant Reflections. At a high level, participants reported that the combination of a familiar environment (all had some prior experience with Google Sheets) and the tool’s simplistic design made them feel comfortable after a short walkthrough. When prompted to complete discovery activities such as determining who had made a change, in all but a few cases it was the first instinct of the participant to go straight to the Log, suggesting that it was understood by participants as a locus information about the dataset. For example, DW 5 described the tool as being, “...*actually self-explanatory; it’s very easy to use.*” In the rest of this section, we describe the more nuanced experiences and reflections of participants, ranging from their existing documentation practices (prior to using Datum Fieldnotes) to what about the tool made it easier or harder to engage in their data work.

Documentation practices status quo. The problems participants had were consistent with what we hypothesized they would be when designing the tool, particularly regarding data documentation and version control. They included documentation disorganization (P5), repeating manual work (DW 5, DW 6, DW 3, DW 2), spending time understanding documentation of others, and translation getting lost (P2). Documentation practices varied between data workers. Some participants used external READMEs (e.g., separate Word documents) or other documents external to the spreadsheet environment in order to record high level changes to a dataset (DW 1, DW 2, DW 6). Difference within use of READMEs varied as well; sometimes they were full records of dataset changes and other times they were merely highlights to be shared with a client (DW 6). Others relied on keeping backup copies of the spreadsheet (e.g., P2), while yet others (e.g., P6) regularly changed their documentation methods depending on the needs of individual clients. Datum Fieldnotes creates a unified, accessible record; while more prolonged testing is needed, all participants reported that they felt the tool made their documentation process easier.

Enabling enhanced communication, upsetting documentation color schematics. The additional “Notes” feature of the tool provided useful communication avenues for participants. Users described how having records of changes and notes, with P3 saying it “*would have saved so much time*” when collaborating with others on a dataset. A few participants (DW 2, DW 4, P6) compared our notes to the built in Google Sheets comments, showing initial confusion over the benefit of their additional inclusion (suggesting a need for us to better differentiate the two), but came to find throughout our interview activities how these two methods of added documentation differ and can be used in conjunction. Most participants (DW 1, DW 3, DW 4, DW 5, DW 6, P2, P3) described struggling with version control in collaborative spreadsheet-based work. Prior to Datum

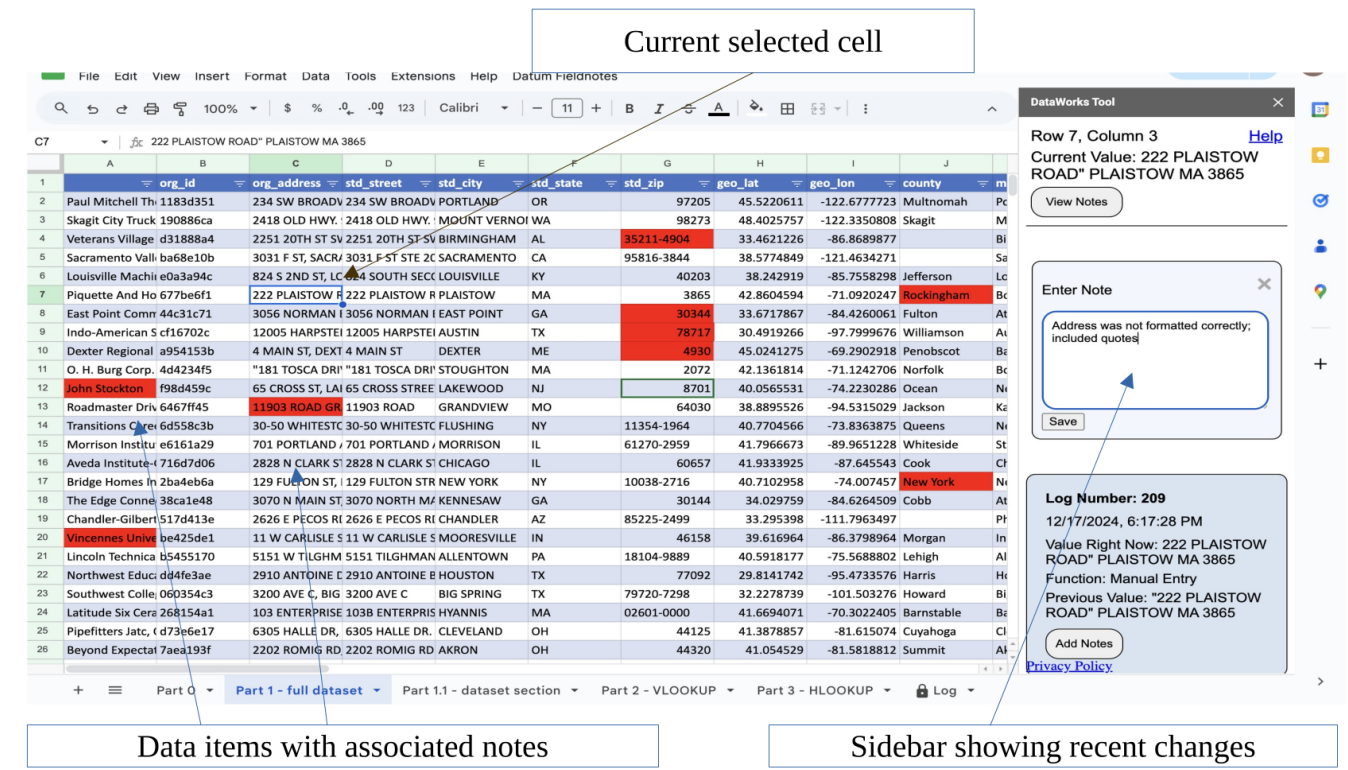


Figure 2: Example dataset with sidebar showing prior edits to the selected cell.

Fieldnotes, they relied on ad-hoc methods of color coding, notes or comments features native to their spreadsheet software, and some participants maintained external central reference points. In particular, the color used to highlight a cell indicated a concern type, e.g., question, data type, or responsible team member.

Self-reflection, growth, and learning in the workplace. Several participants remarked on the Log and Notes' potential as a running "note-to-self" that would be helpful in remembering and keeping track of not just the transformations being enacted on the dataset, but how and why these changes were being made. This matters for CNP data workers who may be learning on the job (particularly volunteer data workers), with P1 explaining, "That's how we learn." However, our findings revealed challenges in utilizing basic tool functions. Many participants struggle with basic tool functions such as viewing and highlighting notes, understanding and utilizing the log due to limited experience with data work and software difference such as Google Sheets and Excel. Participants (DW 2, DW 5, DW 6, DW 7) struggled differentiating in track changes. Particularly DW 6 and DW 7 were confused about using notes to track changes opting to use the log instead, highlighting a need for clearer documentation and support around note functionality. Several participants (P1, DW 2, DW 3, DW 5, DW 6, DW 7) struggled to understand the log's purpose and track changes effectively. Difficulties included identifying the type of change (user, filter, manual), finding the specific filter used, and understanding how users were identified in the log. Users DW 3 and P4 specifically had a hard time

understanding the purpose of the sidebar and finding information. Participants discovered functions incrementally, and some users wanted to know how to view notes. DW 3, for example, found the filter recreation task particularly applicable to their experience with more collaborative work and on-job learning, because they could reference the Log to find and copy a filter or function they needed as (instead of writing it from scratch).

Multiple participants (DW 2, DW 3, DW 5) were unfamiliar with advanced formulas—or their formatting in Google Sheets, though the underlying concepts may not be new—and identified ways that the structure of the tool could better support them. For example, P2 struggled to use filters, as applied by their imaginary co-worker, because the logging of filters did not include the "=" sign that prefaces filtering operations. DW 4 echoed this, describing the struggle of learning a new operation during work, but then not having repeated opportunity to regularly apply it, and subsequently forgetting it. Participants DW 2, DW 3, DW 5, DW 6, DW 7, and P1 also expressed confusion in the value of version control, particularly as enacted by the tool, suggesting the need for more support of tool users learning new data concepts, not just new data transformation functions. These findings underscore the need for tool documentation and support catering to diverse data skill levels. Clearer guidance on note-taking, log utilization, and basic functionalities will help CNP data workers to learn, collaborate effectively, and use the tools full potential for self-reflection and growth.

5 Discussion: Learnings for Developing CNP Data Tools

In response to the user testing, we developed a second version of the tool. These changes include 1) enhancement of data transparency and user-centric design, 2) further logging and documentation features, 3) support for installing and learning to use the tool (and resultant paradata). However, we focus on the the last of these, and its importance for making tools that are sustainable for CNP workers. The overarching finding was that, yes, our tool was usable for CNP data workers after a short instruction, but to make the next step, we needed to better support them in continuing to develop their rich contextualization analysis practices. Namely, throughout our analysis, we found that the benefits offered by the tool could only be enacted when participants had ready access to robust documentation and refresher material. In the CNP context, we argue creating these materials is actually the more important follow-up contribution, given the fluctuation in data worker population and skills, than technical improvements on the tool itself.

As a result, we developed an extensive online curriculum [web-site link removed for review], ranging from simple guides on how to install Datum Fieldnotes and employ its core features, to a series of videos on how to perform analysis on the paradata recorded to the log. To cater to different review methods, we incorporated videos and images alongside text, making it easier for users to grasp the concepts. Recognizing the challenges posed by software differences and high turn over the documentation provides concise instructions, enabling users with varying software experience to quickly learn and apply the techniques to their work. The curriculum starts with simple guides on how to install and set up Datum Fieldnotes in the “User Setup” and “Activating Datum Fieldnotes” sections offering straightforward installation and activation instructions so users can easily access and start using the tool. Topics include lessons like “Filtering the Log Sheet” and “Checking if formulas have been applied consistently”.

These videos are to guide essential data analysis techniques, supporting users in developing their data skills. Other videos demonstrate how to perform analysis on the Log’s paradata record. For example, “Exploring/Querying the Log” provides detailed explanations and examples of how to use filters, sort data, and understand user contributions, addressing the difficulties encountered by participants. These paradata tutorials are meant not only to help orient CNP data workers to these advanced features but to help them position themselves as dataset auteurs. For example, one of the videos covers how to visualize user contributions over time, while another shows how to plot the range of edits over time, which, in many cases, can demonstrate the depth of experience of a dataset contributor.

From this research-through-design process, our findings broadly point to multiple opportunities for design in tandem with CNP data workers, who are yet to receive a commensurate amount of the CHI community’s attention for the data work that they do. Unlike data workers in private sector corporations that have larger budgets and capacities for specialized software, as related work has demonstrated, the ecosystem for those new tool creation and adoption is much different for CNP organizations. We additionally contribute several implications for design.

First, the variance data in work backgrounds of our participants suggests the need to develop tools for CNPs that have both graphical interfaces (such as the Datum Fieldnotes’ sidebar interface) for new to data work, to minimize the burden of adoption, particularly in the case of domain experts who are secondarily performing data work [3, 34].

Second, based on the range of data types our participants were working with—ranging from textual, to numerical, and manual calculations vs imports from tools and sensors—we should consider ways that spreadsheets and their add-ons, as a sustainable software for CNPs, can be adapted to support annotation of additional kinds of data, for example, images.

Third, the security and privacy of CNP data remains a space rife with tensions between different kinds of data and data workers. For some of the participants we talked to, being able to anonymize or lock down their datasets (while still making them usable and annotatable within the organization) was a primary concern, particularly for organizations with clients experiencing conditions with social stigma (e.g., homelessness or health conditions). Other participants wanted to be able to share out their annotations (as public paradata) and wanted to know how Datum Fieldnotes would or would not support that functionality. Additional work is needed on our part to develop a version of the tool that will satisfy the needs across the security and privacy spectrum presented by our participants, with our tool version two presenting some initial improvements. While Datum Fieldnotes is one example of the privacy implications of CNP data work, our observations highlight a key structural problem: CNP data workers need custom, value-sensitive software and tools, but do not have the budgets to support it.

6 Limitations, Future Work, and Conclusion

This work-in-progress has two limitations. First, as we have existing relationships with our participants, we are aware that they might have felt compelled to provide us with positive feedback. In order to preempt this, we structured all recruiting materials and interview questions to make it clear we wanted to elicit critical feedback, in part out of a desire to make the tool usable for our community partners, and we made a point of explicitly asking for features that were uncomfortable or difficult to use, as well as desired additional features. Second, as with any qualitative work and user testing, our sample size is not exhaustive and does not reflect all experiences; however, given the range of data practitioners (organization type) and format of data work, we spoke with, our insights point towards multiple avenues of future work, particularly related to data privacy and sharing tools for CNP data work.

Our contribution in this late-breaking-work is the design and deployment of a Google Sheets add-on to offload the documentation of individual data transactions and re-position data workers as dataset auditors. Complemented by a suite of user testing, we demonstrate the difficulty of designing with and for CNP data workers and identify underappreciated tensions in the design of new data tools for an often-overlooked, but critical, population of data workers.

Acknowledgments

We are grateful to our participants for sharing their time and expertise with us. This work is supported by NSF #1951818 DataWorks: Building Smart Community Capacity and a Kapor Foundation Dissertation Fellowship. Any opinions, findings and conclusions or recommendations expressed in this material do not necessarily reflect the views of NSF or Kapor Foundation.

References

- [1] Philip E. Agre. 2014. Toward a Critical Technical Practice: Lessons Learned in Trying to Reform AI. In *Social Science, Technical Systems, and Cooperative Work* (0 ed.), Geoffrey Bowker, Geoffrey Bowker, Susan Leigh Star, Les Gasser, and William Turner (Eds.). Psychology Press. <https://doi.org/10.4324/9781315805849>
- [2] Lehn M. Benjamin, Amy Volda, and Chris Bopp. 2018. Policy fields, data systems, and the performance of nonprofit human service organizations. *Human Service Organizations: Management, Leadership & Governance* 42, 2 (March 2018), 185–204. <https://doi.org/10.1080/23303131.2017.1422072> Publisher: Routledge _eprint: <https://doi.org/10.1080/23303131.2017.1422072>
- [3] Chris Bopp, Ellie Harmon, and Amy Volda. 2017. Disempowered by Data: Non-profits, Social Enterprises, and the Consequences of Data-Driven Work. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. ACM, Denver Colorado USA, 3608–3619. <https://doi.org/10.1145/3025453.3025694>
- [4] Karen L. Boyd. 2021. Datasheets for Datasets help ML Engineers Notice and Understand Ethical Issues in Training Data. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW2 (Oct. 2021), 1–27. <https://doi.org/10.1145/3479582>
- [5] Virginia Braun and Victoria Clarke. 2021. *Thematic Analysis*. Item ID: _n0 Publication Title: 978-1-4739-5324-6.
- [6] Shiva Darian, Aarjav Chauhan, Ricky Marton, Janet Ruppert, Kathleen Anderson, Ryan Clune, Madeline Cupchak, Max Gannett, Joel Holton, Elizabeth Kamas, Jason Kibozi-Yocka, Devin Mauro-Gallegos, Simon Naylor, Meghan O'Malley, Mehul Patel, Jack Sandberg, Troy Siegler, Ryan Tate, Abigail Temtim, Samantha Whaley, and Amy Volda. 2023. Enacting Data Feminism in Advocacy Data Work. *Proc. ACM Hum.-Comput. Interact.* 7, CSCW1 (April 2023), 47:1–47:28. <https://doi.org/10.1145/3579480>
- [7] Tim Davies and Mark Frank. 2013. ‘There’s no such thing as raw data’. Exploring the socio-technical life of a government dataset. *Proceedings of the 5th Annual ACM Web Science Conference* (2013). https://www.academia.edu/70581019/There_s_no_such_thing_as_raw_data_Exploring_the_socio_technical_life_of_a_government_dataset
- [8] Rosalind Edwards and Janet Holland. 2013. *What is qualitative interviewing?* Bloomsbury, London : New Delhi. OCLC: ocn855705441.
- [9] Sheena Erete, Emily Ryou, Geoff Smith, Khristina Marie Fasset, and Sarah Duda. 2016. Storytelling with Data: Examining the Use of Data by Non-Profit Organizations. In *Proceedings of the 19th ACM Conference on Computer-Supported Cooperative Work & Social Computing (CSCW '16)*. Association for Computing Machinery, New York, NY, USA, 1273–1283. <https://doi.org/10.1145/2818048.2820068>
- [10] Aristeia Fotopoulou. 2021. Conceptualising critical data literacies for civil society organisations: agency, care, and social responsibility. *Information, Communication & Society* 24, 11 (Aug. 2021), 1640–1657. <https://doi.org/10.1080/1369118X.2020.1716041> Publisher: Routledge _eprint: <https://doi.org/10.1080/1369118X.2020.1716041>
- [11] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé Iii, and Kate Crawford. 2021. Datasheets for datasets. *Commun. ACM* 64, 12 (Dec. 2021), 86–92. <https://doi.org/10.1145/3458723>
- [12] Mara Gorli, Davide Nicolini, and Giuseppe Scaratti. 2015. Reflexivity in practice: Tools and conditions for developing organizational authorship. *Human Relations* 68, 8 (Aug. 2015), 1347–1375. <https://doi.org/10.1177/0018726714556156> Publisher: SAGE Publications Ltd.
- [13] Jonathan Gray, Carolin Gerlitz, and Liliana Bounegru. 2018. Data infrastructure literacy. *Big Data & Society* 5, 2 (July 2018), 2053951718786316. <https://doi.org/10.1177/2053951718786316> Publisher: SAGE Publications Ltd.
- [14] Thomas A. Grossman, Vijay Mehrotra, and Özgür Özlük. 2007. Lessons from Mission-Critical Spreadsheets. *Communications of the Association for Information Systems* 20, 1 (Dec. 2007). <https://doi.org/10.17705/1CAIS.02060>
- [15] Ellie Harmon, Chris Bopp, and Amy Volda. 2017. The Design Fictions of Philanthropic IT: Stuck Between an Imperfect Present and an Impossible Future. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems (CHI '17)*. Association for Computing Machinery, New York, NY, USA, 7015–7028. <https://doi.org/10.1145/3025453.3025650>
- [16] Holtzblatt Karen and Jones Sandra. 1993. Contextual Inquiry: A Participatory Technique for System Design. In *Participatory Design*. CRC Press. Num Pages: 34.
- [17] Rob Kitchin and Tracey Lauriault. 2014. Towards Critical Data Studies: Charting and Unpacking Data Assemblages and Their Work. <https://papers.ssrn.com/abstract=2474112>
- [18] Zachary J. Kunicki, Nicholas S. Zambrotta, Marie C. Tate, Angela R. Surrusco, Megan M. Risi, and Lisa L. Harlow. 2019. Keep Your Stats in the Cloud! Evaluating the Use of Google Sheets to Teach Quantitative Methods. *Journal of Statistics Education* 27, 3 (Sept. 2019), 188–197. <https://doi.org/10.1080/10691898.2019.1665485> Publisher: Taylor & Francis _eprint: <https://doi.org/10.1080/10691898.2019.1665485>
- [19] Seolha Lee and Paul Dourish. 2024. Reconfiguring Data Relations: Institutional Dynamics around Data in Local Governance. *Proc. ACM Hum.-Comput. Interact.* 8, CSCW2 (Nov. 2024), 420:1–420:28. <https://doi.org/10.1145/3686959>
- [20] Sabina Leonelli. 2016. Locating ethics in data science: responsibility and accountability in global and distributed knowledge production systems. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 374, 2083 (Dec. 2016), 20160122. <https://doi.org/10.1098/rsta.2016.0122>
- [21] Anthony McCosker, Xiaofang Yao, Kath Albury, Alexia Maddox, Jane Farmer, and Julia Stoyanovich. 2022. Developing data capability with non-profit organisations using participatory methods. *Big Data & Society* 9, 1 (Jan. 2022), 20539517221099882. <https://doi.org/10.1177/20539517221099882> Publisher: SAGE Publications Ltd.
- [22] Dan McQuillan. 2022. *Resisting AI: an anti-fascist approach to artificial intelligence*. Bristol University Press, Bristol, UK. OCLC: on1328026349.
- [23] Anna Mills and Nate Angell. 2025. Are We Tripping? The Mirage of AI Hallucinations. <https://doi.org/10.2139/ssrn.5127162>
- [24] Margaret Mitchell, Alexandra Sasha Luccioni, Nathan Lambert, Marissa Gerchick, Angelina McMillan-Major, Ezinwanne Ozoani, Nazneen Rajani, Tristan Thrush, Yacine Jernite, and Douwe Kiela. 2022. Measuring Data. <https://doi.org/10.48550/arXiv.2212.05129> arXiv:2212.05129 [cs].
- [25] National Council of Nonprofits. [n.d.]. (Mis)Understanding Overhead. <https://www.councilofnonprofits.org/running-nonprofit/administration-and-financial-management/misunderstanding-overhead>
- [26] Kim Nault, Umar Ruhi, and Omer Livvarcin. [n.d.]. Exploring the Applications & Challenges of Data Analytics in Non-Profit Organizations. *AMCIS 2020 Session 8* ([n.d.]).
- [27] Aare Puusaar, Ian G. Johnson, Kyle Montague, Philip James, and Peter Wright. 2018. Making Open Data Work for Civic Advocacy. *Proc. ACM Hum.-Comput. Interact.* 2, CSCW (Nov. 2018), 143:1–143:20. <https://doi.org/10.1145/3274412>
- [28] Annabel Rothschild, Amanda Meng, Carl DiSalvo, Britney Johnson, Ben Rydal Shapiro, and Betsy DiSalvo. 2022. Interrogating Data Work as a Community of Practice. *Proceedings of the ACM on Human-Computer Interaction* 6, CSCW2 (Nov. 2022), 1–28. <https://doi.org/10.1145/3555198>
- [29] Sidney Joseph Shapiro and Vivian Oystriick. 2018. Three Steps Toward Sustainability: Spreadsheets as a Data-Analysis System for Non-Profit Organizations. *Canadian Journal of Program Evaluation* 33, 2 (Sept. 2018), 247–257. <https://doi.org/10.3138/cjpe.31157> Publisher: University of Toronto Press.
- [30] Caroline Sindors, Rainbow Unicorn, Cade Diehm, and Ian Ardouni Fumat. 2020. Technically Responsible Knowledge (TKS). <https://carolinesindors.com/trk/>
- [31] Pieter Jan Stappers and Elisa Giaccardi. 2024. Research through Design. In *The Encyclopedia of Human-Computer Interaction* (2nd ed.). Interaction Design Foundation. <https://www.interaction-design.org/literature/book/the-encyclopedia-of-human-computer-interaction-2nd-ed/research-through-design>
- [32] Ciaran B. Trace and James A. Hodges. 2024. The Role of Paradata in Algorithmic Accountability. In *Perspectives on Paradata: Research and Practice of Documenting Process Knowledge*, Isto Huvila, Lisa Andersson, and Olle Sköld (Eds.). Springer International Publishing, Cham, 197–213. https://doi.org/10.1007/978-3-031-53946-6_11
- [33] Amy Volda. 2011. Shapeshifters in the voluntary sector: exploring the human-centered-computing challenges of nonprofit organizations. *Interactions* 18, 6 (Nov. 2011), 27–31. <https://doi.org/10.1145/2029976.2029985>
- [34] Amy Volda, Ellie Harmon, and Ban Al-Ani. 2011. Homebrew databases: complexities of everyday information management in nonprofit organizations. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '11)*. Association for Computing Machinery, New York, NY, USA, 915–924. <https://doi.org/10.1145/1978942.1979078>
- [35] Rachel Wells. 2024. Google Sheets Vs. Microsoft Excel—Which Is Better? *Forbes* (May 2024). <https://www.forbes.com/sites/rachelwells/2024/05/01/google-sheets-vs-microsoft-excel-which-is-better/> Section: Careers.
- [36] Hannah West and Gina Green. 2008. Because Excel will Mind Me! The State of Constituent Data Management in Small Nonprofit Organizations. *AMCIS 2008 Proceedings* (Jan. 2008). <https://aisel.aisnet.org/amcis2008/336>
- [37] Shwu-Ing Wu and Jr-Ming Hung. 2008. A performance evaluation model of CRM on non-profit organisations. *Total Quality Management & Business Excellence* 19, 4 (April 2008), 321–342. <https://doi.org/10.1080/14783360701591978> Publisher: Routledge.
- [38] Jonathan Zong and J. Nathan Matias. 2023. Data Refusal From Below: A Framework for Understanding, Evaluating, and Envisioning Refusal as Design. *ACM Journal on Responsible Computing* (Oct. 2023). <https://doi.org/10.1145/3630107> Just Accepted.